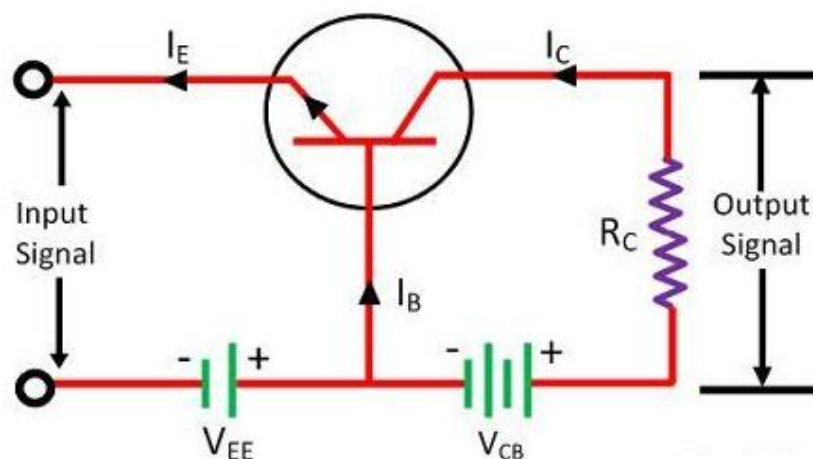


22GE004 - BASICS OF ELECTRONICS ENGINEERING

UNIT III - Signal Amplification using BJT

1. Transistor as an amplifier

The transistor raises the strength of a weak signal and hence acts as an amplifier. The transistor amplifier circuit is shown in the figure below. The transistor has three terminals namely emitter, base and collector. The emitter and base of the transistor are connected in forward bias and the collector-base region is in reverse bias. The forward bias means the P-region of the transistor is connected to the positive terminal of the supply and the negative region is connected to the N-terminal and in reverse bias just opposite of it has occurred.



The input signal or weak signal is applied across the emitter-base and the output is obtained from the load resistor R_C which is connected in the collector circuit. The DC voltage V_{EE} is applied to the input circuit along with the input signal to achieve the amplification. The DC voltage V_{EE} keeps the emitter-base junction under the forward biased condition regardless of the polarity of the input signal and is known as a bias voltage.

When a weak signal is applied to the input, a small change in signal voltage causes a change in emitter current (or we can say a change of 0.1V in signal voltage causes a change of 1mA in the emitter current) because the input circuit has very low resistance. This change is almost the same in collector current because of the transistor action.

In the collector circuit, a load resistor R_C of high value is connected. When collector current flows through such a high resistance, it produces a large voltage drop across it. Thus, a weak signal (0.1V) applied to the input circuit appears in the amplified form (10V) in the collector circuit.

Applications of Transistor as an Amplifier

Transistor working as an amplifier has various advantages and applications in the field of electronics and communication. They are

1. It can be used in long-distance communication because the intensity of the signal obtained at the output will be high.
2. In the radio signals amplification, these transistor amplifiers are utilized.
3. The amplifiers play a prominent role in wireless communication.
4. The amplification of the signals using transistors as amplifiers can be utilized in the FM signals broadcasting.
5. In the optical Fiber Communication also these types of amplifiers are utilized.
6. The basic application of this transistor amplifier is as an audio amplifier and this is used in our day to day activities that we come across.

DISCUSSION QUESTIONS:

1. Identify the different types of transistors commonly used as amplifiers.
2. Outline the steps involved in designing a transistor amplifier circuit.
3. Identify the key components involved in a transistor amplifier circuit.